

University Database Analysis Report

SQL Server | Python | Power BI | Education Analytics

Prepared by	Yasir Awad
Institution	MEC Academy
Project Type	End-to-End Database Analytics Project

A professional report documenting the project overview, database structure, analytical workflow, Power BI dashboard pages, key findings, data risks, and recommendations.

Executive Summary

Project objective. This report presents an end-to-end analysis of a normalized university database using SQL Server, Python, and Power BI. The analysis was designed to answer academic and operational questions related to student performance, course capacity, teaching quality, prerequisite compliance, scheduling, and data quality.

Business value. The project demonstrates how a university can transform structured academic data into actionable insights for academic planning, capacity optimization, teaching quality monitoring, and evidence-based decision-making.

591 Total Students	108 Total Courses	41 Total Professors	28 Total Classrooms	18 Total Majors
87.36% Pass Rate	73.90 Avg Final Score	1.90 Avg GPA	1,499 Total Evaluations	72.39% Response Rate

Summary of findings. The project identified a strong academic performance base, with approximately 68% of students achieving grades between A and C. Computer Science was highlighted as the top-performing major, while prerequisite compliance issues were identified for 203 students. Teaching quality showed a moderate positive relationship with student final scores, and capacity analysis revealed opportunities to rebalance course and classroom utilization.

Project Overview and Business Questions

Project Scope

The project analyzes a normalized university database consisting of academic, administrative, scheduling, enrollment, assessment, and evaluation data. The deliverables include SQL analysis, Python analysis, Power BI dashboards, and an executive summary.

Core Deliverables

- SQL queries for academic and operational data retrieval.
- Python analysis for validation, feature engineering, and visualization.
- Power BI report pages for student success, capacity, and teaching quality.
- Executive insights and practical recommendations.

Key Questions Addressed

- Which majors and student cohorts perform best academically?
- How are students distributed across GPA and grade levels?
- Are students enrolling in courses without completing prerequisites?
- Which courses and classrooms are under-utilized or overloaded?
- How does teaching evaluation relate to student final scores?
- Are there scheduling conflicts across classrooms, professors, or students?
- What KPIs should academic management monitor regularly?

The project uses a structured analytics workflow that starts with database understanding, continues through SQL and Python validation, and ends with business-facing Power BI dashboards designed for decision-makers.

Database Structure

The database follows a normalized relational model with interconnected entities across academic structure, course delivery, scheduling, assessments, enrollments, and evaluations. The structure supports detailed analysis of student outcomes, course utilization, teaching quality, and operational constraints.

Area	Core Entities	Business Use
Academic Structure	Departments, Majors, Professors, Students	Track academic hierarchy, student cohorts, and staff distribution.
Course Catalog	Courses, Prerequisites	Analyze course offerings and prerequisite compliance.
Course Delivery	Course Offerings, Enrollments	Measure enrollment, performance, capacity, and utilization.
Scheduling	Timeslots, Classrooms, Offering Timeslots	Detect scheduling conflicts and classroom utilization issues.
Assessment	Assessment Components, Student Component Scores	Recompute final scores and validate assessment data.
Feedback	Evaluations, Evaluation Summary	Analyze teaching quality and evaluation response rates.

Data quality considerations. The project includes checks for missing values, invalid ranges, duplicates, foreign key integrity, assessment weight sums, evaluation aggregation consistency, and scheduling conflicts.

Methodology and Analytics Workflow

Stage	Activities	Outputs
SQL Server	Loaded CSV tables, queried enrollments, capacity, prerequisites, grades, teaching evaluations, and scheduling conflicts.	Labeled SQL query outputs and result samples.
Python	Validated tables, enforced data types, recomputed academic metrics, generated analytical aggregates, and created visualizations.	Notebook analysis, quality checks, charts, and reconciliation outputs.
Power BI	Built a model and dashboard pages for student success, course capacity, teaching quality, and university-level KPIs.	Interactive dashboard with slicers, KPI cards, charts, and tables.
Executive Reporting	Summarized findings, data risks, and recommendations for academic decision-making.	Professional report and presentation-ready insights.

The workflow reflects a complete end-to-end analytics process: database understanding, data validation, metric recomputation, dashboard development, and executive communication.

Power BI Dashboard Structure

Dashboard Page	Purpose	Main Focus
Student Success	Analyze GPA trends, grade distribution, student performance, and cohort patterns.	Academic outcomes and student performance monitoring.
Course & Capacity	Evaluate total courses, classrooms, majors, enrollments, capacity, and utilization.	Academic planning and resource utilization.
Teaching Quality	Review evaluation response rates, professor evaluation scores, and faculty-level relationships.	Teaching feedback, instructional quality, and evaluation reliability.

Each dashboard page was designed to translate technical database outputs into clear KPIs, charts, and comparisons that can support academic management decisions.

Key Findings

Academic Performance • Computer Science was identified as the top-performing major based on average GPA. • Approximately 68% of students achieved grades between A and C. • Average GPA was around 1.90, while average final score was approximately 73.90.	Academic Compliance • 203 students were found to have enrolled in courses without completing required prerequisites. • This indicates a need for stronger registration validation and prerequisite enforcement. • Prerequisite checks can reduce academic progression risks and improve course readiness.	Teaching and Operations • Teaching evaluation scores showed a moderate positive relationship with final student scores. • Capacity and utilization analysis highlighted opportunities to rebalance course and classroom usage. • No double-booked classrooms were identified, while scheduling checks remain important for operational control.
--	--	--

The findings show that the university database can be used not only for descriptive reporting, but also for operational control, academic quality monitoring, and policy improvement.

Data Risks and Recommendations

Area	Issue / Opportunity	Recommended Action
Prerequisites	Some students were enrolled without completing required prerequisites.	Implement automated prerequisite validation during registration.
Capacity	Some courses and classrooms may be under-utilized or overloaded.	Rebalance course sections and classroom allocation using utilization KPIs.
Data Quality	Missing or inconsistent values appeared in a few enrollment and assessment records.	Apply periodic reconciliation checks across assessments, enrollments, and evaluation summaries.
Teaching Feedback	Some courses may have incomplete or low evaluation response coverage.	Improve feedback participation and monitor response rates by course and term.
Management Reporting	Academic KPIs need continuous monitoring.	Use Power BI dashboards for recurring review of GPA, pass rate, utilization, and evaluation performance.

These recommendations are designed to improve data governance, academic planning, teaching quality monitoring, and registration compliance.

Student Success Dashboard

This page focuses on GPA trends, grade distribution, pass rate, average final score, average GPA, and student performance by major and gender.

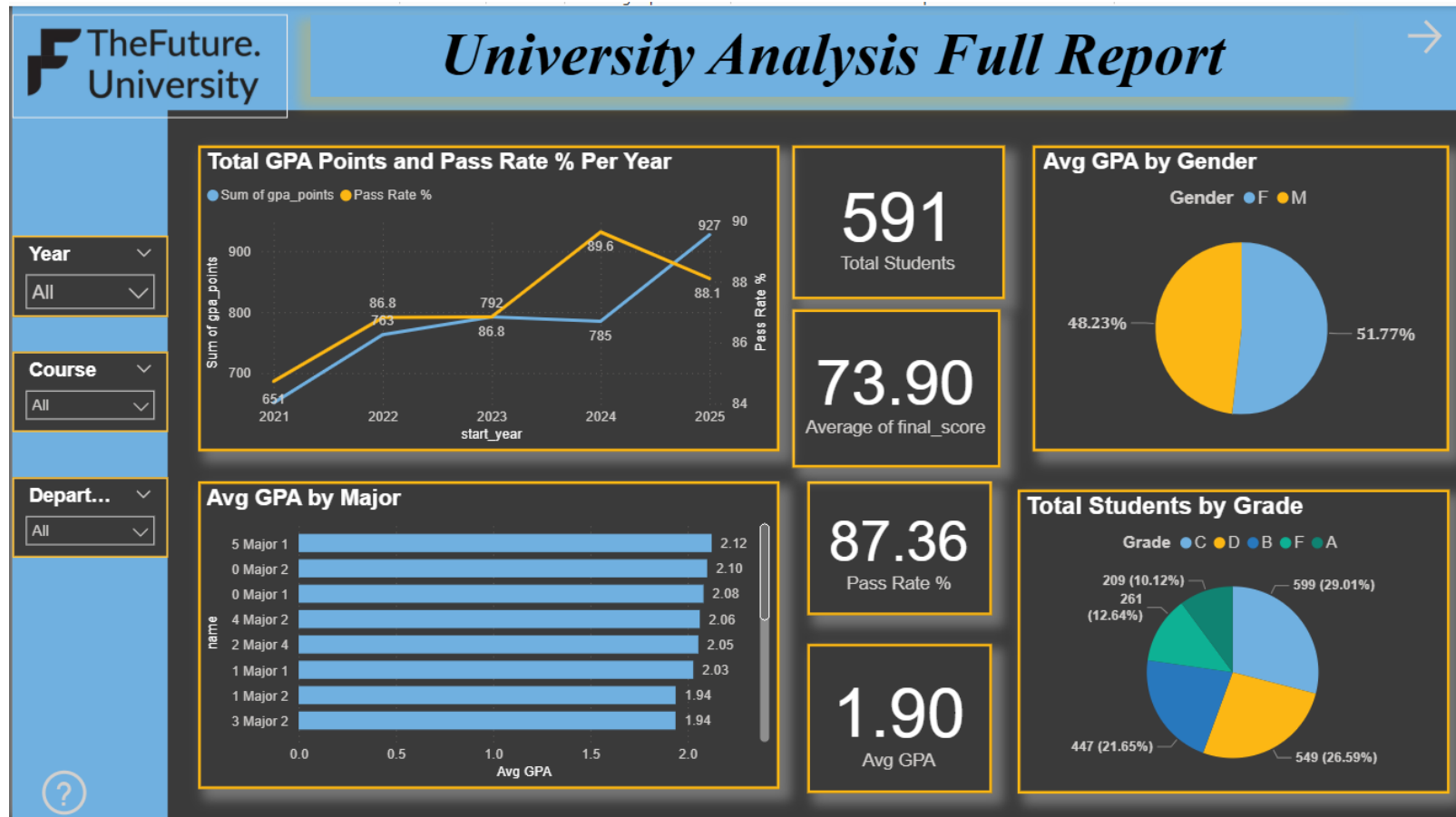


Figure: Student Success Dashboard

Course and Capacity Dashboard

This page evaluates course offerings, classroom capacity, utilization, enrollments by faculty, and enrollment versus capacity by course.

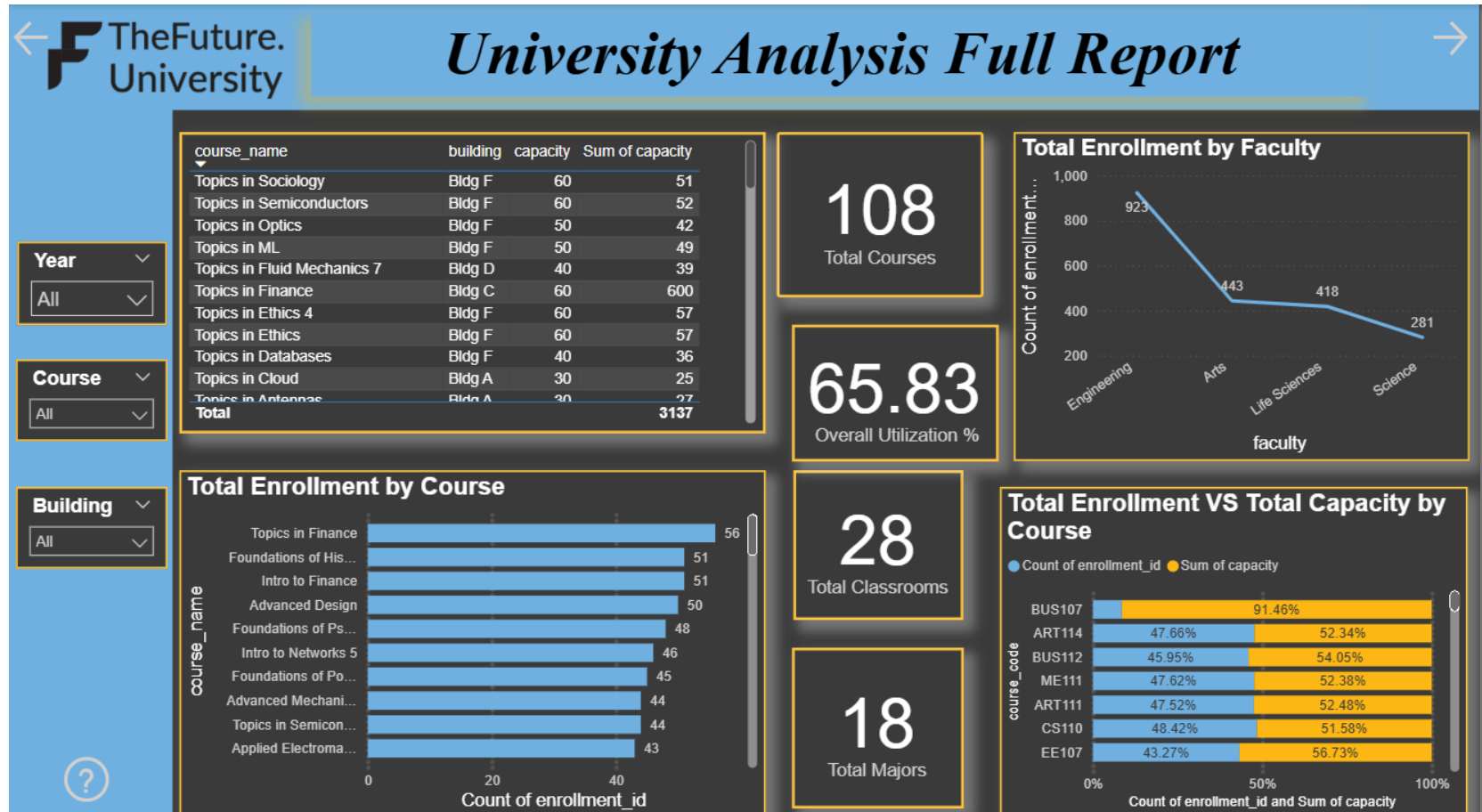


Figure: Course and Capacity Dashboard

Teaching Quality Dashboard

This page analyzes evaluation volume, response rate, average evaluation scores, professor-level evaluations, and enrollment versus evaluations by faculty.

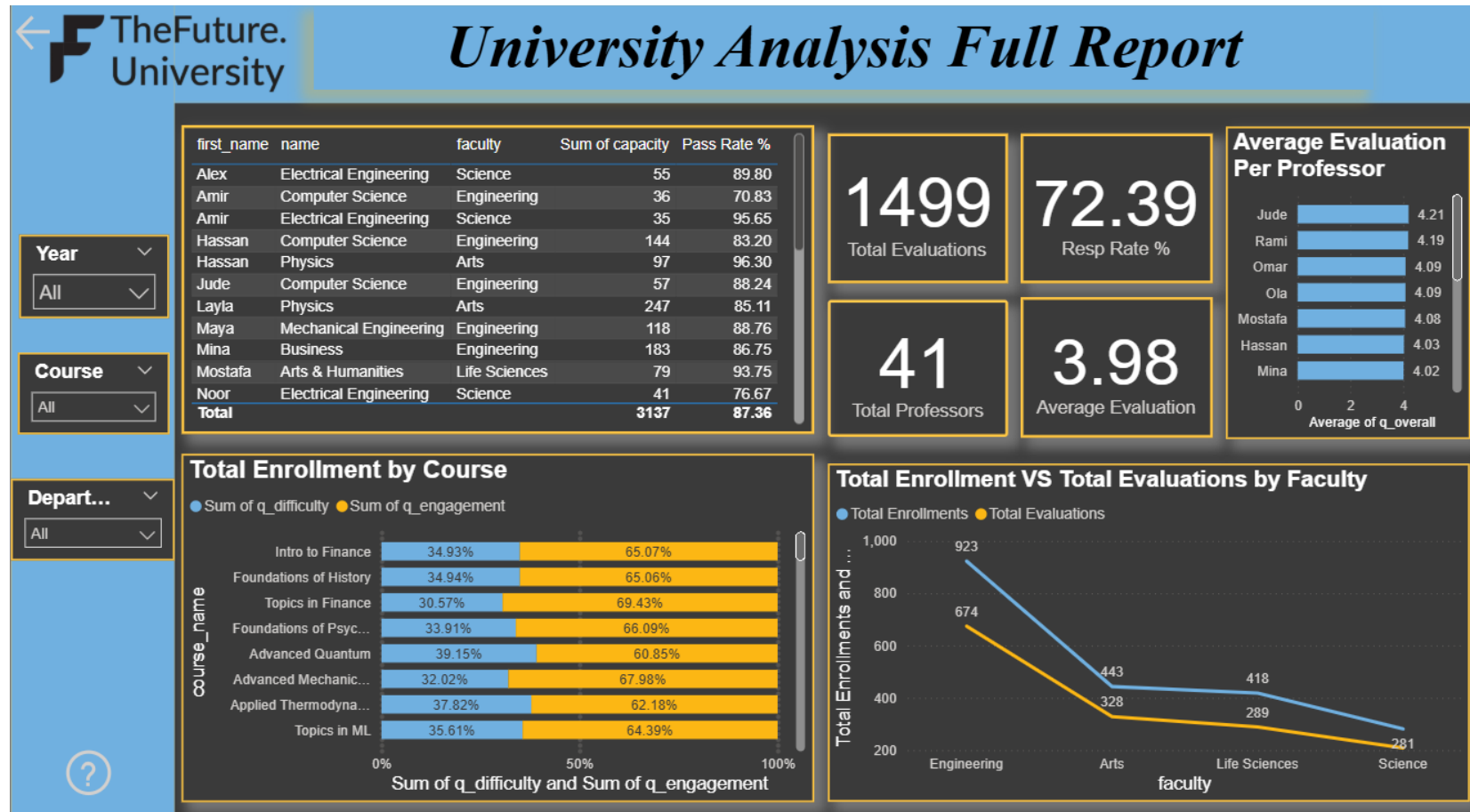


Figure: Teaching Quality Dashboard

Technical Workflow Evidence

The project also included Python-based data validation, SQL database connectivity, data type inspection, summary checks, and analytical outputs. The screenshots below illustrate parts of the notebook workflow used to connect to the database, inspect tables, and validate academic metrics.

```
Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')

# Display all outputs from each cell
from IPython.core.interactiveshell import InteractiveShell
InteractiveShell.ast_node_interactivity = "all"

# Formatting the output
pd.options.display.float_format = '{:.2f}'.format

[1]
```

```
Connect to Database System

from sqlalchemy import create_engine
import pandas as pd
server = "DESKTOP-83P462" # From SELECT @@SERVERNAME
database = "University_DB" # your database name

[2]
```

Python setup and database connection

```
# cell 2
summary = []
for name, df in tables.items():
    summary.append({
        'table': name,
        'rows': len(df),
        'cols': df.shape[1],
        'null_percent_sample %': (df.isnull().mean().max() * 100).round(2),
        'dtypes': ', '.join([f"{t}" for c, t in df.dtypes.items()][5:]) + ('...' if df.shape[1] > 5 else '')
    })
pd.DataFrame(summary).set_index('table')
```

	rows	cols	null_percent_sample %	dtypes
table				
departments	6	5	0.00	department_idint64, dept_codeobject, nameob...
majors	18	4	0.00	major_idint64, department_idint64, major_cod...
professors	41	6	0.00	professor_idint64, department_idint64, first...
students	600	5	0.00	student_idint64, major_idint64, start_yeari...
courses	108	5	0.00	course_idint64, department_idint64, course_c...
prerequisites	10	2	0.00	course_idint64, prereq_course_idint64
classrooms	28	4	0.00	classroom_idint64, buildingobject, room_num...
timeslots	25	4	0.00	timeslot_idint64, day_of_weekobject, start_t...
course_offerings	61	7	0.00	offering_idint64, course_idint64, professor_...
offering_timeslots	61	3	0.00	offering_idint64, timeslot_idint64, classroo...
enrollments	2065	6	0.15	enrollment_idint64, student_idint64, offeri...
assessment_components	217	6	0.00	component_idint64, offering_idint64, nameob...

Table-level summary and data quality inspection

1 - Ingests all CSVs, enforces types, validates FK integrity, checks rules

```
departments = pd.read_sql("SELECT * FROM departments", engine)
majors = pd.read_sql("SELECT * FROM majors", engine)
professors = pd.read_sql("SELECT * FROM professors", engine)
students = pd.read_sql("SELECT * FROM students", engine)
courses = pd.read_sql("SELECT * FROM courses", engine)
prerequisites = pd.read_sql("SELECT * FROM prerequisites", engine)
classrooms = pd.read_sql("SELECT * FROM classrooms", engine)
timeslots = pd.read_sql("SELECT * FROM timeslots", engine)
course_offerings = pd.read_sql("SELECT * FROM course_offerings", engine)
offering_timeslots = pd.read_sql("SELECT * FROM offering_timeslots", engine)
enrollments = pd.read_sql("SELECT * FROM enrollments", engine)
assessment_components = pd.read_sql("SELECT * FROM assessment_components", engine)
student_component_scores = pd.read_sql("SELECT * FROM student_component_scores", engine)
evaluations = pd.read_sql("SELECT * FROM evaluations", engine)
evaluation_summary = pd.read_sql("SELECT * FROM evaluation_summary", engine)
```

Loading tables from the database

```
enrollments.drop(columns=['grade_new', 'gpa_new'], inplace=True)

print(enrollments['gpa_points'].describe())
print(enrollments['grade'].value_counts())
```

```
[12]
```

```
... count    2065.00
     mean         1.90
     std         1.18
     min         0.00
     25%         1.00
     50%         2.00
     75%         3.00
     max         4.00
Name: gpa_points, dtype: float64
grade
C    599
D    549
B    447
F    261
A    209
Name: count, dtype: int64
```

GPA statistics and grade distribution output

Project Outcome and Repository Files

This project demonstrates the ability to work across the full data analytics workflow, from database understanding and SQL querying to Python validation and Power BI dashboard development. It shows how academic institutions can use analytics to monitor performance, improve planning, and support evidence-based decisions.

Folder / File	Purpose
README.md	Project documentation and dashboard preview.
powerbi/university-analysis-dashboard.pbix	Power BI dashboard file.
docs/university-database-analysis-report.pdf	Professional project report.
docs/university-database-analysis-presentation.pptx	Project presentation deck.
images/*.png	Dashboard screenshots and Python workflow screenshots for GitHub and portfolio preview.
notebooks/analysis.ipynb	Python analysis notebook, if added to the repository.
sql/answers.sql	SQL queries and analysis outputs, if added to the repository.

Prepared by Yasir Awad - Data Analyst | Business Intelligence | Energy & Operations Analytics

Email: yasir.petro.analytics@outlook.com | GitHub: github.com/Yasir101-hi | LinkedIn: linkedin.com/in/yasirawad